

FullPull : A Stretchable UI to Input Pulling Strength on Touch Surfaces

Tatsuya Maeda
Aoyama Gakuin University
tatsuya.maeda@x-lab.team

Kodai Ito
Aoyama Gakuin University
kodai.ito@it.aoyama.ac.jp

Keita Kuwayama
Osaka University
kuwayama.keita@ist.osaka-u.ac.jp

Kazuyuki Fujita
Tohoku University
k-fujita@riec.tohoku.ac.jp

Yuichi Itoh
Aoyama Gakuin University
ito@it.aoyama.ac.jp

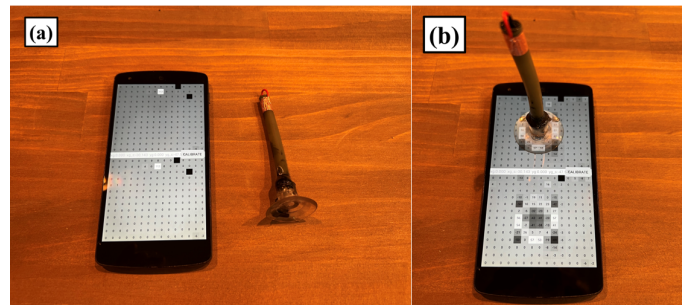


Figure 1: (a) The rubber tube interface (FullPull). (b) Capacitance image when the interface attached to surface

ABSTRACT

Touch surfaces are used as input interfaces for many devices such as smartphones, tablets, and smartwatches. However, the flexibility of the input surface is low, and the possible input operations are limited to planar ones such as touch and swipe. In contrast, in the field of HCI, there has been much research on increasing the number of input interactions by attaching augmented devices with various physical characteristics to the touch surface. However, most of these interactions are limited to operations where pressure is applied to the input surface. In this study, we propose *FullPull*, which consists of a rubber tube filled with conductive ink and a suction cup to attach the rubber tube to the surface. FullPull allows users to input pulling depth and strength on the touch surface. We implemented a prototype FullPull device which can be attached to an existing capacitive touch surface and can be pulled by a user. We then evaluated the accuracy of tensile strength estimation of the implemented device. The results showed that the outflow current value when stretched could be classified into four tensile strength levels.

KEYWORDS

HCI, Capacitive touch surface, Outflow current value, Tensile, Rubber

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1 INTRODUCTION

Touch surfaces such as smartphones, tablets, and smartwatches have become remarkably popular and indispensable in our daily lives. Many touch surfaces employ projected capacitance [1] as a touch recognition method. However, touch surfaces in general use today are inflexible and planar, and the operations that can be performed on the surface are limited to input on a flat surface and be one direction, such as touch and swipe. Therefore, we cannot handle three-dimensional information and operations such as pushing, pulling, pinching, and bending directly on the surface. In the field of human-computer interaction (HCI), there has been a lot of research to tackle these problems by adding recognizable input operations and expanding the input area using general touch surfaces. Many studies have been conducted to attach extended devices with various physical shapes and characteristics to the touch surface and to enable input by utilizing the physical characteristics of these devices that are not possible with rigid and flat surfaces. There has enabled extension input areas, buttons with force feedback during operation, isometric joysticks, pressure input, bending input, and shear force input [2][3][4][5][6][7]. In contrast, we considered to detect forces in the direction away from the touch surface as inputs. In this study, we focus on "pulling" operations, which are general

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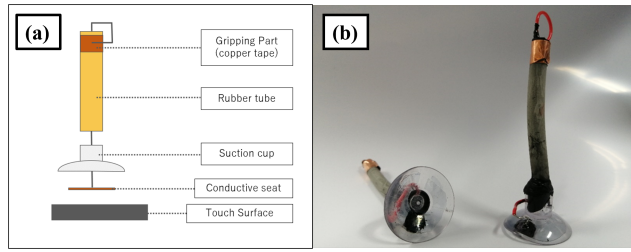


Figure 2: (a) Overview of the FullPull Device. (b) Implemented FullPull prototype by the rubber tube filled with conductive ink

actions in daily life, such as pulling the door. We propose an interface named *FullPull* that can be attached to existing projected capacitive touch surfaces to enable pulling operations in the front direction perpendicular to the input plane. The device consists of a combination of rubber to enable pulling and a suction cup to fix the interface to the touch surface. And the capacitive touch surface sensor detects changes in the electrical characteristics (resistance) of the interface during pulling to recognize the pulling operation.

2 A PROTOTYPE OF FULLPULL DEVICE

A FullPull device is fixed to the surface using suction cups for tensile operation. The touch surface in popular use today detects touch using the outflow current value measured by a sensor inside the surface. Changes in the impedance of the object between the surface and the GND affect the value of the outflow current value measured by the sensors [4]. The device we propose increases resistance as tensile strength increases. Combining these two features enables tensile manipulation on the interface. An overview of the interface is shown in Figure 2 (a). The rubber tube interface consists of a rubber tube, and suction cup to attach the interface to the surface. The rubber tube is filled with conductive ink and is electrically connected from the copper-tape-wrapped gripping part to the surface of the suction cup. The rubber tube has an inner diameter of 5 mm and an outer diameter of 7 mm, and the length of the conductive ink-filled part is 5 cm. We inserted conducting wires from both ends of the rubber tube, sealed the holes with rubber solvent-based adhesive, and the rubber tube was combined with a suction cup and glued together. The suction cups were 3 cm in diameter. The completed prototype is shown in Figure 2 (b).

3 EVALUATION

We conducted experiment using the implemented prototype device to investigate the relationship between the tensile strength of the interface and the outflow current value of the surface. The experimental setup is shown in Figure 1. The touch surface used in the experiment is LG Nexus5. The device was attached to the center of the screen with a suction cup, and pinched the gripping part of the device and pulled in 0.5 cm increments up to a maximum of 5.0 cm. We recorded the outflow current values when the rubber tube interface was pulled perpendicular to the screen. Tensile operation was performed 10 times for each length of tension, for a total of 100 times. Figure 3 shows the results of the experiment. As

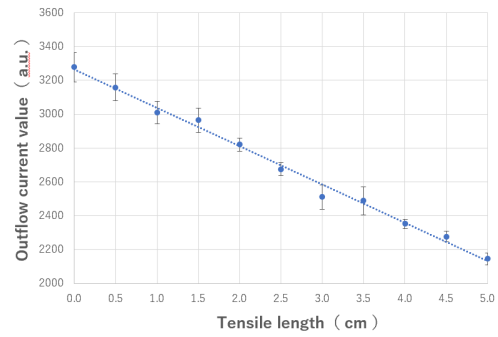


Figure 3: Relationship between device tensile length and outflow current value

shown in the figure, the outflow current value decreases almost linearly as the tensile length increases. Multiple comparisons using the Tukey method showed that the averages were significantly different among the outflow current values for tensile lengths with a difference of 1.5 cm or more. Therefore, if the implemented interface is used to pull up to 5.0 cm, four levels of tensile strength estimation (0 cm, 1.5 cm, 3.0 cm, and 4.5 cm) can be performed.

4 POSSIBLE APPLICATIONS

We describe possible applications for a FullPull device. It allows for depth and strength input of a flat display, making three-dimensional actions more intuitive and providing the reaction force of rubber. As a gaming device, it can be used as a controller such as plucking strings, like a bow. Since it can recognize the tensile strength, it is possible to adjust the speed and power. As a work tool, it can be applied as a metaphor for copying and pasting. For example, some information can be stored and restored where it is needed, and if the device is assigned an ID, pasting to different screens can be done intuitively. Alternatively, it may help with tasks that require the input of subtle intensity changes, such as drawing or multimedia editing.

5 CONCLUSION AND FUTURE WORK

In this study, we proposed a rubber tube interface named FullPull, which can be fixed to a touch surface and can be operated by tension. We confirmed that the interface could estimate the tensile strength at four levels. However, the capacitance change when a person touches the touch surface is affected by the GND state of the surface and the person, the touch state (pressure and ground area), and the human body impedance. The evaluation experiment in this study was conducted by one person, it is necessary to consider the influence of the human body impedance in the future. In addition, we will create specific applications using the rubber tube interface and investigate the possibility of recognizing operations other than pulling, such as "pinching" and "twisting".

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